MIL-T-36321A
26 August 1970
SUPERSEDING
MIL-T-36321 (DM)
2November 1964

MILITARY SPECIFICATION

TRACTION APPARATUS, BONE FRACTURE

This specification is mandatory for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 This specification covers one type of portable bone fracture traction apparatus, suitable for use in applying traction to arms and legs of a patient during orthopedic surgical procedures involving the application of a cast.

2. APPLICABLE DOCUMENTS.

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-A-200/3A	Aluminum Alloy Bar, Rod, Shapes, Tube and Wire Extruded 2024.
QQ-A-596	Aluminum Alloy Permanent and Semi-Permanent Mold Castings.
QQ-B-613	Brass, Leaded and Non-Leaded; Flat Products (Plate, Bar, Sheet, and Strip).
QQ-S-698	Steel, Sheet and Strip, Low-Carbon.
QQ-T-830	Tube, Steel, (Carbon, Mechanical; Seamless and Welded).
TT-E-491	Enamel, Gloss, Synthetic (for Metal and Wood Hospital Furniture).
TT-E-527	Enamel, Alkyd, Lustreless.
TT-P-636	Primer Coating, Alkyd, Wood and Ferrous Metal.

FSC 6530

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CCC-C-419	Cloth, Duck, Cotton, Unbleached, Plied-Yarns, Army and Numbered.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

Federal

Fed. Std. No. 66	Steel: Chemical Composition and Hardenability.
Fed. Std. No. 595	Colors.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

DRAWING

Defense Personnel Support Center (DPSC)

20486	Traction Apparatus, Bone Fracture, Portable.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

3. REQUIREMENTS

3.1 Material. Material used in the fabrication of portable traction apparatus shall conform to the applicable requirements indicated in Table I. and shown in Figure I. Commercial tolerances shall apply to all gauges.

3.1.1 Lumber. Lumber used in the fabrication of the carrying case (see 3.2.2) shall be beech, birch, maple or oak. The lumber shall be well seasoned and kiln dried, and free from decay, shakes, checks, cracks, or other defects which may affect the strength, wear or serviceability of the case.

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TABLE I - MATERIALS

Part No.	Nomenclature	Req'd	Material	Specification	Dimensions (inches)
1	Base	1	Aluminum casting, alloy 319	QQ-A-596	(see drawing)
-	Base insert	1	MT 1010 or 1020, CDSR, seamless or welded tubing	QQ-T-830	1-3/8 OD by 0.120 (No. 11, B.W. gage) wall by 5-3/4 long.
-	Base pin	1	AISI 303 corrosion-resisting steel rod	Fed Std No. 66	1/4 diameter by 2-1/2 long.
2	Headrest	1	Aluminum casting, alloy 319	QQ-A-596	(see drawing)
3	Shoulder rest	1	Aluminum casting, alloy 319	QQ-A-596	(see drawing)
4	Outside extension tube	2	MT 1010 or 1020, CDSR, seamless or welded tubing	QQ-T-830	1-1/16 OD by 0.083 (No. 14, B.W. gage) wall by 30-1/8 long.
-	Tube insert	2	MT 1010 or 1020, CDSR, seamless or welded tubing	QQ-T-830	7/8 OD by 0.109 (No. 12, B.W. gage) wall by 10-5/8 long.
5	Inside extension tube	2	MT 1010 or 1020, CDSR, seamless or welded tubing	QQ-T-830	7/8 OD by 0.095 (No. 13, B.W. gage) wall by 30 long.
6	Traction screw guide tube	2	MT 1010 or 1020, CDSR, seamless or welded tubing	QQ-T-830	7/8 OD by 0.109 (No. 12, B.W. gage) wall by 2 long.
-	Guide pin	2	AISI 1018 or 1020 cold rolled steel rod	comm'l	1/8 diameter by 3/4 long.

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TABLE I - MATERIALS

Part No.	Nomenclature	Req'd	Material	Specification	Dimensions (inches)
7	Wingnut	2	Brass	----------	(see drawing)
8	Sacral rest	1	Aluminum casting, alloy 319	QQ-A-596	(see drawing)
-	Locating pin	1	AISI 303 corrosion-resisting steel rod	Fed Std No. 66	3/16 diameter by 3/4 long.
9	Headrest support block	1	AISI 1018 cold rolled steel	comm'l	7/8 square by 6-7/8 long.
10	Foot support rod	4	AISI 1018 cold-rolled steel rod	comm'l	3/8 diameter by 10-5/8 long.
11	Headrest support bar	2	AISI 1018 cold-rolled steel rod	comm'l	5/8 diameter by 6-1/8 long.
12	Headrest slider bar	2	AISI 1018 cold-rolled steel rod	comm'l	3/8 diameter by 20-1/8 long.
13	Leg support bar	2	AISI 1018 cold-drawn steel rod	comm'l	1/2 diameter by 24 long.
14	Lower headrest bar	2	AISI 1018 cold-rolled steel rod	comm'l	1/2 diameter by 27-1/8 long.
15	Traction screw	2	AISI 1018 cold-rolled steel rod	comm'l	5/8 diameter by 8-19/32 long.
-	Retaining pin	2	AISI 303 corrosion-resisting steel rod	Fed Std No. 66	1/8 diameter by 1-5/16 long.
16	Upright tube	1	MT 1010 or 1020, CDSR, seamless or welded tubing	QQ-T-830	1-1/8 OD by 0.095 (No. 13, B.W. gage) wall by 19-1/4 long.

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TABLE I - MATERIALS

Part No.	Nomenclature	Req'd	Material	Specification	Dimensions (inches)
17	Upright tube crosspiece	1	MT 1010 or 1020, CDSR, seamless or welded tubing	QQ-T-830	1-1/8 OD by 0.095 (No. 13, B.W. gage) wall by 8-7/8 long.
--	End plug	2	AISI 1010, CRDQ, steel sheet	QQ-S-698	1 diameter by 1/16 thick (see drawing).
18	Perineal post	1	Aluminum casting, alloy 319	QQ-A-596	(see drawing)
--	Locating pin	1	AISI 303 corrosion-resisting steel rod	Fed Std No. 66	3/16 diameter by 3/4 long.
19	Extension rod lock screw	2	AISI 303 corrosion-resisting steel rod	Fed Std No. 66	1/2 diameter by 3-7/8 long.
20	Arm extension bar	1	MT 1010 or 1020, CDSR, seamless or welded tubing	QQ-T-830	1-1/16 diameter by 0.083 (No. 14, B.W. gage) wall by 30-1/8 long.
21	Swivel block	4	Aluminum casting, alloy 319	QQ-A-596	(see drawing)
22	Swivel head	2	SAE 2024 aluminum alloy	QQ-A-200/3A	1-1/4 diameter sphere
23	Support Block insert	2	Free machining brass	QQ-B-613	5/8 OD by 2-1/8 long.
24	Body sling	1	No. 12 White duck unbleached canvas	CCC-C-419	(see drawing)
25	Ankle hitch assembly	2	No. 12 White duck unbleached canvas	CCC-C-419	(see drawing)

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3.1.2 Hardware. All screws, bolts, nuts, washers and other hardware required for proper assembly and use of the traction apparatus shall be in accordance with Table I and Drawing 20486.

3.2 Construction.

3.2.1 Traction apparatus. The portable traction apparatus shall be as shown in figure 1, Drawing 20486, and as specified herein. The traction apparatus shall be so constructed that all moving parts move freely and the perineal post is easily removable. The apparatus, when tested as specified in 4.4.1, shall be capable of withstanding a leg traction of 400 pounds without bending, distortion or any other failure.

3.2.2 Carrying case. A sturdy, rigid, wood, carrying case shall be provided. The case shall be fabricated of not less than 5/8 inch thick lumber. When panels are made of two or more widths, they shall be neatly and closely tongue-and grooved and glued. All joints shall be shaped (such as tongue and groove, lock corners, or dovetail) and glued. Glue shall be a water-resistant casein type, prepared and applied according to the glue manufacturer's recommendations. All exposed surfaces of wood shall be evenly and smoothly dressed. The case shall be assembled to true 90 degree angles and the cover shall fit down on the body with not more than a 1/32 inch opening at any point. The inside of the cover shall be not less than 2 inches deep. The case shall be completely assembled with flathead wood screws, countersunk flush with the exterior wood surface. Screws shall be spaced on 3 inch centers. The case shall be blocked so that all components of the traction apparatus are securely held in place. The case shall be furnished with a steel carrying handle, steel hinges, and hasps.

3.3 Finish.

3.3.1 Steel. All carbon steel sheets, rods and tubing shall be chromium plated over nickel over copper. All corrosion-resisting steel shall be given a No. 4 commercial finish.

3.3.2 Brass. All brass parts shall be plated with chromium over nickel over copper. All brass castings shall be smoothly polished and buffed prior to plating.

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3.3.3 Carrying case. All hardware, except screws, before assembly to the case, shall be given a phosphate treatment followed by a coating of synthetic primer conforming to TT-P-636. The inside and outside of the case, before any hardware is attached, shall be given one coat of synthetic semi-gloss enamel conforming to TT-E-527. After complete assembly of the hardware to the case, the entire inside and outside surfaces, including the exterior surfaces of the hardware, shall be given another coat of olive drab enamel. The color shall conform to color X24087 of Fed. Std. 595 (see 6.3).

3.3.4 Cast aluminum. Cast aluminum parts shall be smoothly polished, buffed and given two coats of enamel conforming to TT-E-491. The color shall be green conforming to color shade No. 14241 of Fed. Std. 595. The parts shall be baked after each coat to yield a smooth hard finish.

3.4 Identification marking. Each case shall have a metal nameplate, securely and firmly attached bearing the following information:

Federal stock number.

Name or registered trademark of the manufacturer.

Contract or purchase order number.

Letters U.S.

Marking shall be distinct, legible and permanent. Marking secured by adhesives shall not be acceptable.

3.5 Service data. The contractor shall enclose, with each unit, two copies each of one or more booklets which contain complete instructions for installation, operation, maintenance, and a list of component parts. Each part of the unit other than common fasteners, shall be identified in the service data and parts list by the manufacturer's part number. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the contracting officer, via the cognizant Government inspector, two preliminary copies of his proposed service data, for review and comment by the Defense Personnel Support Center. The preliminary service data shall be submitted sufficiently early to permit adequate review by the Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of required revisions and corrections in the final service data without delaying delivery under the contract or purchase order. In addition, the contractor shall mail 15 copies of the approved service data to the contracting officer for distribution.

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3.6 Workmanship. Workmanship shall be first class throughout. The traction apparatus and its carrying case shall be free from defects which detract from their appearance or which may impair their serviceability. All metal parts shall be free from burs and sharp edges and corners. All brazed parts shall be sound, and shall be cleaned.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection: Inspection, as used in this specification, is defined as both examination (such as visual, and auditory, investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the traction apparatus or carrying case may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the FSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Classification of Inspections:

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and Table II. The unit of product for examination shall be the traction apparatus and carrying case.

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Table II

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	S-3	1.0
Total (major and minor combined) defects	S-3	4.0
For dimensional examination	S-3	2.5

4.2.2 For tests. Sampling for leg traction test shall be in accordance with inspection level S-3 of MIL-STD-105. The acceptance number shall be zero regardless of lot size.

4.3 Examination. The traction apparatus shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the classification of defects shown in table III below.

Table III

Categories	Defects
Major	
101	Apparatus or carrying case not complete with all required component parts.
102	Metal parts not free of cracks, ruptures, or deformed sections.
103	Exposed metal surface not free of burs or sharpness.
104	Brazed joints not sound or not free of cracks, flux, discoloration, or spatter.
105	Moving parts cannot be assembled; or not free of jamming; or do not adjust freely.
106	Threaded parts stripped, missing, incomplete.
107	Wood or carrying case not free of cracks, splinters, or deformed surfaces.
108	Plated surfaces of apparatus or painted surfaces of case not free of blisters, and peeled or uncoated areas.
Minor	
201	Castings not buffed and polished.
202	Identification marking missing, incomplete, or illegible.
203	Service data (in duplicate) not included or not as specified.
204	Apparatus, carrying case not clean; not free of dirt, oil, grease.
205	Any screw in case not flush with wood surface.
206	Wood of carrying case not free of knots.
207	Exposed surfaces of wood of carrying case not free of roughness.

* Examination shall not be restricted to the classified possible defects listed above.

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4.3.2 Dimensional examination. The apparatus shall be examined for defects in dimensions. Any dimension not within the tolerances specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Leg traction. A force of 400 pounds shall be exerted between the swivel block of one foot attachment assembly and the junction of the sacral rest and the perineal post. The test shall be repeated between swivel block of the other foot attachment assembly and the top junction of sacral rest and perineal post.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One traction apparatus, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit packaging. Exterior surfaces of the carrying case shall be protected by wrapping with corrugated fiberboard. Wrapping shall be secured by cord or tape. Work shall be accomplished in such a manner as to insure against marring, scratching, or damage to the unit.

5.1.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Each unit shall be packed in a exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3. Closure and strapping shall be as specified in the appendix of the applicable box specification.

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5.2.1.2 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.2 Level B.

5.2.2.1 Exterior container. Each unit shall be packed in a exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Use: For applying traction to the arms or legs of a patient during orthopedic surgical procedures involving application of a cast.

6.2 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.

(b) Selection of applicable level of packaging and packing (see 5.1 and 5.2).

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6.3 Availability of color standards. The color standards of the color referenced in 3.3.3 are available from:

Commanding General
Aberdeen Proving Ground,
Attn: Coating and Chemical laboratory.

6.4 This specification covers the following item appearing in the Federal Supply Catalog:

FSN	Item
6530-709-9400	Traction Apparatus, Bone Fracture, Portable

6.5 This specification does not include all types, classes, grades, and sizes of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, which are normally procured to meet Military requirements.

6.6 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue."

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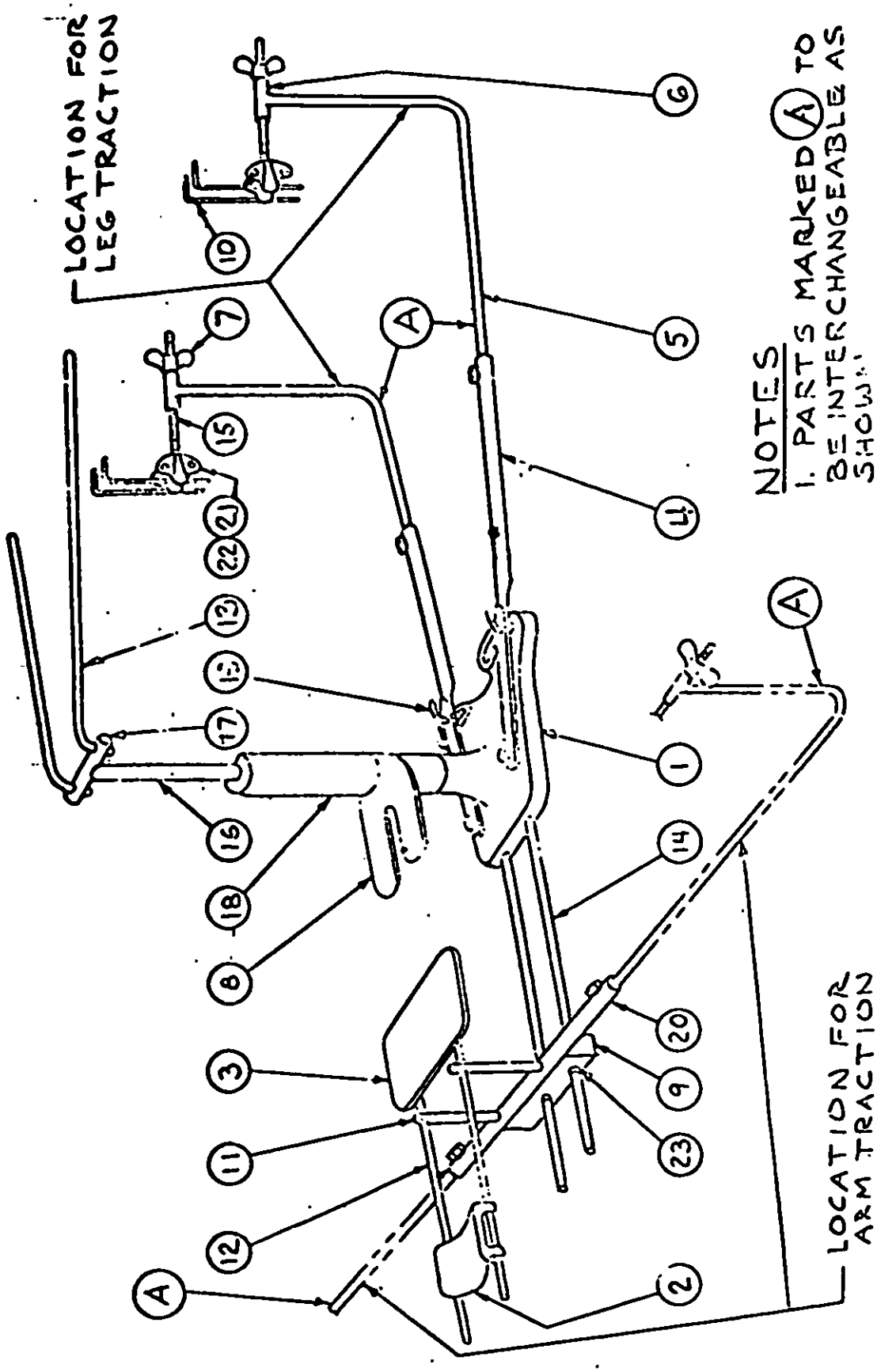


FIGURE 1

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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

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